ic Dataset Analysis and Data Cleaning - 4 01:34

rovided with the Titanic dataset containing information about passengers
anic. Your task is to write Python code to answer the following questions
the dataset.

et the number of survivors by gender (Sex).
et the number of non-survivors by gender (Sex).
et the number of survivors by embarkation location (Embarked_S).
et the number of non-survivors by embarkation location (Embarked_S).
lculate the percentage of children (Age < 18) who survived.
lculate the percentage of adults (Age >= 18) who survived.
et the median age of survivors.
et the median age of non-survivors.
et the median fare of survivors.
et the median fare of non-survivors.

ic dataset contains columns as shown below,

S ur	P	N		A	Si	P	Ti	F	C	E m

Explorer titanicDat...

```
# import pandas as pd
# import numpy as np

# # Load the Titanic dataset
# data = pd.read_csv('Titanic-Dataset.csv')
```

Average time 0.058 s 58.00 ms

Maximum time 0.058 s 58.00 ms

1 out of 1 shown test case(s) pass

Test case 1 58 ms Debug

Expected output	Actual output
female····233	female····233
male·····109	male·····109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male·····468	male·····468
female····81	female····81
Name: Sex, dtype: int64	Name: Sex, dtype: int64
1····217	1····217

Terminal Test cases

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s - series creation and manipulation 00:56

thon program that takes a list of numbers from the user, creates a ries from it, and then calculates the mean of even and odd numbers using the `groupby` and `mean()` operations.

can group the Series based on whether each number is even or odd.

mat:

user should enter a list of numbers separated by space when mpted.

rmat:

program should display the mean of even and odd numbers separately. h mean value should be displayed with a label indicating whether it esponds to even or odd numbers.

e sample test cases for better understanding regarding input and output

Explorer

seriesMa...

```
import pandas as pd

# Take inputs from the user to create a list of num
numbers = list(map(int, input().split()))

```

Average time **0.045 s** 45.50 ms

Maximum time **0.074 s** 74.00 ms

3 out of 3 shown test case(s) pa

3 out of 3 hidden test case(s) pa

Test case 1 74 ms Debug

Expected output	Actual output
1 2 3 4 5 6 7 8 9 10	1 2 3 4 5 6 7 8 9 10
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd 5.0	Odd 5.0
Even 6.0	Even 6.0
dtype: float64	dtype: float64

Terminal Test cases

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onary to dataframe 01:12

ary of lists has been provided to you in the editor. Create a DataFrame dictionary of lists and perform the listed operations, then display the ne before and after each manipulation.

ne DataFrame:

onvert the dictionary to a Pandas DataFrame.

ew row:

ke inputs from the user for the new row data (name, age).

ld the new row to the DataFrame.

splay the DataFrame after adding the new row.

row:

odify a specific row by changing the age. Take the row index and new age lue from the user.

splay the DataFrame after modifying the row.

row:

ke the row index to be deleted from the user.

emove the specified row

Explorer

dataframe...

```
'''import pandas as pd

# Provided dictionary of lists
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
```

Average time	Maximum time
0.234 s	0.234 s
233.50 ms	234.00 ms

1 out of 1 shown test case(s) pass

1 out of 1 hidden test case(s) pass

Test case 1 233 ms Debug

Expected output	Actual output
Original DataFrame:	Original DataFrame:
Name Age	Name Age
0 Alice 25	0 Alice 25
1 Bob 30	1 Bob 30
2 Charlie 35	2 Charlie 35
New name: Susan	New name: Susan

Terminal Test cases

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ent Information 00:40

rogram to read a text file containing student information (name, age, and
sing Pandas. Perform the following tasks:

splay the first five rows of the data frame.
alculate the average age of the students (limit the average age up to 2
cimal places).
ter out the students who have a grade above a certain threshold (consider
e threshold grade is **'B'**).

he displayed test cases for better understanding.

le Test Cases +

Explorer

studentin...

```
import pandas as pd

# Read the text file into a DataFrame
file = input()
data = pd.read_csv(file, sep="\s+", header=None,
```

Average time	Maximum time
0.095 s	**0.120 s**
94.50 ms	120.00 ms

1 out of 1 shown test case(s) pass

1 out of 1 hidden test case(s) pass

Test case 1 120 ms Debug

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
...Name Age Grade	...Name Age Grade
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A

Terminal Test cases

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th with the Highest Total Sales 06:23

ython program that takes the file name of a CSV file as input, reads the
performs the following operations:
e CSV file contains the columns: `Date`, `Product`, `Quantity`,
`rice`, and `City`.
roup the data by Month and calculate the total sales for each month.
nd the month with the highest total sales and display it.
so, display the total sales for the best month.

Data:

```
oduct,Quantity,Price,City
-01,Product A,5,20,New York
-01,Product B,3,15,Los Angeles
-02,Product A,7,20,New York
-02,Product C,4,30,Chicago
-03,Product B,2,15,Chicago
-03,Product A,8,20,Los Angeles
-04,Product C,6,30,New York
-04,Product B,5,15,Los Angeles
-05,Product A,3,20,Chicago
-05,Product C,10,30,Los Angeles
```

Explorer

monthFor...

```
# import pandas as pd

# # Prompt the user for the file name
# file_name = input()
```

Average time	Maximum time
0.027 s	0.031 s
27.33 ms	31.00 ms

1 out of 1 shown test case(s) pass

2 out of 2 hidden test case(s) pass

Test case 1 31 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: $1210.00	Total sales: $1210.00

Terminal Test cases

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Selling Product 00:37

ython program that takes the file name of a CSV file as input, reads the
performs the following operations:
e CSV file contains the columns: `Date`, `Product`, `Quantity`,
`rice`, and `City`.
nd the product that sold the most in terms of quantity sold.
splay the product that sold the most and the total quantity sold for that
oduct.

Data:

```
oduct,Quantity,Price,City
-01,Product A,5,20,New York
-01,Product B,3,15,Los Angeles
-02,Product A,7,20,New York
-02,Product C,4,30,Chicago
-03,Product B,2,15,Chicago
-03,Product A,8,20,Los Angeles
-04,Product C,6,30,New York
-04,Product B,5,15,Los Angeles
-05,Product A,3,20,Chicago
-05,Product C,10,30,Los Angeles
```

Explorer

monthFor...

```
# import pandas as pd

# # Prompt the user for the file name
# file_name = input()
```

Average time	Maximum time
0.026 s	0.029 s
25.67 ms	29.00 ms

1 out of 1 shown test case(s) pass

2 out of 2 hidden test case(s) pass

Test case 1 29 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best selling product: Product A	Best selling product: Product A
Total quantity sold: 23	Total quantity sold: 23

Terminal Test cases

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that Sold the Most Products 00:49

ython program that takes the file name of a CSV file as input, reads the
performs the following operations:
e CSV file contains the columns: `Date`, `Product`, `Quantity`,
ice, and `City`.
oup the data by `City` and calculate the total quantity of products sold for
ch city.
nd the city that sold the most products (based on the total quantity sold).

Data:

```
oduct,Quantity,Price,City
-01,Product A,5,20,New York
-01,Product B,3,15,Los Angeles
-02,Product A,7,20,New York
-02,Product C,4,30,Chicago
-03,Product B,2,15,Chicago
-03,Product A,8,20,Los Angeles
-04,Product C,6,30,New York
-04,Product B,5,15,Los Angeles
-05,Product A,3,20,Chicago
-05,Product C,10,30,Los Angeles
```

Explorer

monthFor...

```python
import pandas as pd

# Prompt the user for the file name
file_name = input()

```

Average time	Maximum time
0.039 s	0.066 s
39.33 ms	66.00 ms

1 out of 1 shown test case(s) pass

2 out of 2 hidden test case(s) pass

Test case 1 30 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
City sold the most products: Los Angel es	City sold the most products: Los An es

Terminal Test cases

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t Frequently Sold Product Pairs 05:40

ython program that takes the file name of a CSV file as input, reads the
l performs the following operations:
e CSV file contains the following columns: Date, Product, Quantity, Price,
d City.
r each date, find all pairs of products that were sold together (i.e., two
oducts sold on the same date).
utput the product pair/s that was sold most frequently.

Data:

```
oduct,Quantity,Price,City
-01,Product A,5,20,New York
-01,Product B,3,15,Los Angeles
-02,Product A,7,20,New York
-02,Product C,4,30,Chicago
-03,Product B,2,15,Chicago
-03,Product A,8,20,Los Angeles
-04,Product C,6,30,New York
-04,Product B,5,15,Los Angeles
-05,Product A,3,20,Chicago
-05,Product C,10,30,Los Angeles
```

Explorer

frequentl...

```
# import pandas as pd
# from itertools import combinations
# from collections import Counter

# # Prompt user to input the file name
```

Average time 0.039 s 39.33 ms

Maximum time 0.067 s 67.00 ms

1 out of 1 shown test case(s) pass

2 out of 2 hidden test case(s) pass

Test case 1 31 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Product A and Product B: 2 times	Product A and Product B: 2 times
Product A and Product C: 2 times	Product A and Product C: 2 times

Terminal Test cases

Prev Reset Subm

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ic Dataset Analysis and Data Cleaning 01:53

rovided with the Titanic dataset containing information about passengers
anic. Your task is to write Python code to answer the following questions
the dataset. For each question, perform necessary data cleaning,
ations, and calculations as required.

splay the first 5 rows of the dataset.
splay the last 5 rows of the dataset.
et the shape of the dataset (number of rows and columns).
et a summary of the dataset (using `.info()`).
et basic statistics (mean, standard deviation, etc.) of the dataset using
`describe()`.
heck for missing values and display the count of missing values for each
lumn.
l missing values in the 'Age' column with the median age.
l missing values in the 'Embarked' column with the most frequent value
`ode()`).
op the 'Cabin' column due to many missing values.
eate a new column, 'FamilySize' by adding the 'SibSp' and 'Parch'
lumns.

Explorer

titanicDat...

```
# import pandas as pd
# import numpy as np

# # Load the Titanic dataset
# data = pd.read_csv('Titanic-Dataset.csv')
```

Average time **0.209 s** 209.00 ms

Maximum time **0.209 s** 209.00 ms

1 out of 1 shown test case(s) pass

Test case 1 209 ms Debug

Expected output

```
   PassengerId  Survived  Pclass
...     Fare Cabin  Embarked
0            1         0       3
...   7.2500   NaN         S
1            2         1       1
...  71.2833   C85         C
2            3         1       3
...   7.9250   NaN         S
```

Actual output

```
   PassengerId  Survived  Pclass
...     Fare Cabin  Embarked
0            1         0       3
...   7.2500   NaN         S
1            2         1       1
...  71.2833   C85         C
2            3         1       3
...   7.9250   NaN         S
```

Terminal Test cases

Prev Reset Subm

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ic Dataset Analysis and Data Cleaning - 2 00:43

rovided with the Titanic dataset containing information about passengers
anic. Your task is to write Python code to answer the following questions
the dataset.
eate a new column 'IsAlone' which is 1 if the passenger is alone
amilySize = 0), otherwise 0.
onvert the 'Sex' column to numeric values (male: 0, female: 1).
ne-hot encode the 'Embarked' column, dropping the first category.
et the mean age of passengers.
et the median fare of passengers.
et the number of passengers by class.
et the number of passengers by gender.
et the number of passengers by survival status.
alculate the survival rate of passengers.
alculate the survival rate by gender.

ic dataset contains columns as shown below,

S ur	P	N		A	Si	P	Ti	F	C	E m

Explorer titanicDat...

```
# import pandas as pd
# import numpy as np

# # Load the Titanic dataset
# data = pd.read_csv('Titanic-Dataset.csv')
```

Average time 0.041 s 41.00 ms

Maximum time 0.041 s 41.00 ms

1 out of 1 shown test case(s) pass

Test case 1 41 ms Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3····491	3····491
1····216	1····216
2····184	2····184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0····577	0····577

Terminal Test cases

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ic Dataset Analysis and Data Cleaning - 3 00:44

rovided with the Titanic dataset containing information about passengers
anic. Your task is to write Python code to answer the following questions
the dataset.

lculate the survival rate by class.
lculate the survival rate by embarkation location (Embarked_S).
lculate the survival rate by family size (FamilySize).
lculate the survival rate by being alone (IsAlone).
et the average fare by passenger class (Pclass).
et the average age by passenger class (Pclass).
et the average age by survival status (Survived).
et the average fare by survival status (Survived).
et the number of survivors by class (Pclass).
et the number of non-survivors by class (Pclass).

ic dataset contains columns as shown below,

S ur	P	N		A	Si	P	Ti	F	C	E m

Explorer

titanicDat...

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
```

Average time 0.088 s 88.00 ms

Maximum time 0.088 s 88.00 ms

1 out of 1 shown test case(s) pass

Test case 1 88 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.506073	0 0.506073

Terminal Test cases

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